

Ex.no:3	EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Aim:

To perform data cleaning and data transformation on the Daily Google Search Trends US dataset by identifying missing values, handling missing values, removing duplicate records, detecting outliers, and normalizing data using pandas.

Algorithm:

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the Daily Google Search Trends US dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the Daily Google Search Trends US dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows or columns containing any missing value.
7. Use how='all' to remove rows or columns only if all values are missing.
8. Use thresh=N to keep rows or columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
!pip -q install kagglehub

import kagglehub

import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt
```

```

import glob

import os

dataset_path = kagglehub.dataset_download("adrianjuliusaluoch/daily-google-search-trends-us")

csv_files = glob.glob(os.path.join(dataset_path, "**", "*.csv"),
recursive=True)

df = pd.read_csv(csv_files[0])

print("Original dataset shape:", df.shape)

print("Column names:", df.columns.tolist())

print("First 5 records:")

display(df.head())

missing_mask = df.isnull()

print("Missing-value mask:")

display(missing_mask.head())

missing_counts = df.isnull().sum()

print("Missing values per column:")

print(missing_counts)

plt.figure(figsize=(12,5))

sns.barplot(x=missing_counts.index, y=missing_counts.values)

plt.title("Missing Values Count per Column")

plt.xlabel("Columns")

plt.ylabel("Number of Missing Values")

plt.xticks(rotation=45, ha="right")

plt.tight_layout()

plt.show()

from sklearn.impute import SimpleImputer

num_cols = df.select_dtypes(include=np.number).columns.tolist()

cat_cols = df.select_dtypes(include=["object", "category"]).columns.tolist()

if num_cols:

    num_imputer = SimpleImputer(strategy="mean")

    df[num_cols] = num_imputer.fit_transform(df[num_cols])

if cat_cols:

    cat_imputer = SimpleImputer(strategy="most_frequent")

    df[cat_cols] = cat_imputer.fit_transform(df[cat_cols])

print("Missing values after replacement:")

print(df.isnull().sum())

duplicate_count = df.duplicated().sum()

print("Total duplicate records found:", duplicate_count)

```

```

df.drop_duplicates(inplace=True)
df.reset_index(drop=True, inplace=True)
print("Dataset shape after removing duplicates:", df.shape)
numeric_features = df.select_dtypes(include=np.number).columns.tolist()
df_before_outlier_capping = df.copy()
for col in numeric_features:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - (1.5 * IQR)
    upper_bound = Q3 + (1.5 * IQR)
    outliers = ((df[col] < lower_bound) | (df[col] > upper_bound)).sum()
    print(f"\nColumn: {col}")
    print("Q1:", Q1)
    print("Q3:", Q3)
    print("IQR:", IQR)
    print("Lower bound:", lower_bound)
    print("Upper bound:", upper_bound)
    print("Outliers detected:", outliers)
    df[col] = np.where(df[col] < lower_bound, lower_bound, df[col])
    df[col] = np.where(df[col] > upper_bound, upper_bound, df[col])
if numeric_features:
    plot_col = numeric_features[0]
    plt.figure(figsize=(12,5))
    plt.subplot(1,2,1)
    sns.boxplot(y=df_before_outlier_capping[plot_col])
    plt.title("Before Outlier Capping")
    plt.ylabel(plot_col)
    plt.subplot(1,2,2)
    sns.boxplot(y=df[plot_col])
    plt.title("After IQR Outlier Capping")
    plt.ylabel(plot_col)
    plt.tight_layout()
    plt.show()
for col in df.columns:
    if "date" in col.lower() or "time" in col.lower():
        converted = pd.to_datetime(df[col], errors="coerce")

```

```
        if converted.notna().sum() > 0:
            df[col] = converted

string_cols =
df.select_dtypes(include=["object", "category"]).columns.tolist()

for col in string_cols:
    df[col] = df[col].astype(str).str.strip().str.lower()

if string_cols:
    plot_col = string_cols[0]
    top_values = df[plot_col].value_counts().head(10)
    plt.figure(figsize=(10,5))
    sns.barplot(x=top_values.index, y=top_values.values)
    plt.title(f"Top 10 Values - {plot_col}")
    plt.xlabel(plot_col)
    plt.ylabel("Count")
    plt.xticks(rotation=45, ha="right")
    plt.tight_layout()
    plt.show()

print("Final cleaned dataset shape:", df.shape)
print("Remaining missing values:")
print(df.isnull().sum())
print("Final data types:")
print(df.dtypes)
print("First 5 rows of cleaned dataset:")
display(df.head())
```

Output:

```
Downloading to /root/.cache/kagglehub/datasets/adrianjuliusaluoch/daily-google-search-trends-us/194.archive...
100%|██████████| 935k/935k [00:00<00:00, 79.0MB/s]
Extracting files...
Dataset downloaded to:
/root/.cache/kagglehub/datasets/adrianjuliusaluoch/daily-google-search-trends-us/versions/194

CSV file found:
/root/.cache/kagglehub/datasets/adrianjuliusaluoch/daily-google-search-trends-us/versions/194/trending_searches_in_us.csv

Original dataset shape: (31804, 8)

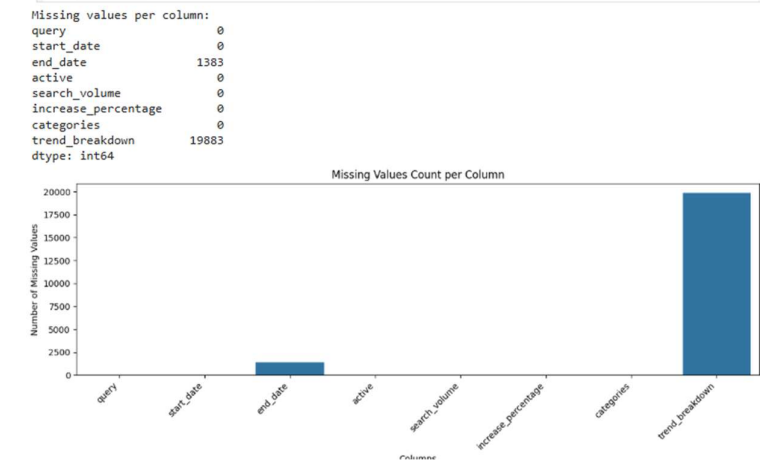
Column names:
['query', 'start_date', 'end_date', 'active', 'search_volume', 'increase_percentage', 'categories', 'trend_breakdown']
```

First 5 records:

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	shedeur sanders	2026-03-30 18:40:00+00:00	NaN	True	200	50	Sports	NaN
1	whoopi goldberg	2026-03-30 18:40:00+00:00	NaN	True	5000	50	Politics	america first award, mike johnson
2	mo williams	2026-03-30 18:00:00+00:00	NaN	True	5000	300	Sports	mason williams basketball, mason williams
3	sloane stephens	2026-03-30 17:50:00+00:00	NaN	True	1000	200	Sports, Health	NaN
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	NaN	True	5000	200	Entertainment	renewed and cancelled tv shows 2026

Missing-value mask (first 5 rows):

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	False	False	True	False	False	False	False	True
1	False	False	True	False	False	False	False	False
2	False	False	True	False	False	False	False	False
3	False	False	True	False	False	False	False	True
4	False	False	True	False	False	False	False	False



```
Numeric columns: ['search_volume', 'increase_percentage']
Categorical columns: ['query', 'start_date', 'end_date', 'categories', 'trend_breakdown']
```

Missing values after replacement:

```
query          0
start_date     0
end_date       0
active         0
search_volume  0
increase_percentage 0
categories     0
trend_breakdown 0
dtype: int64
```

Total duplicate records found: 0

Dataset shape after removing duplicates: (31804, 8)

Numeric features: ['search_volume', 'increase_percentage']

Column: search_volume

```
Q1: 500.0
Q3: 10000.0
IQR: 9500.0
Lower bound: -13750.0
Upper bound: 24250.0
Outliers detected: 3188
```

Column: increase_percentage

```
Q1: 100.0
Q3: 300.0
IQR: 200.0
Lower bound: -200.0
Upper bound: 600.0
Outliers detected: 3099
```

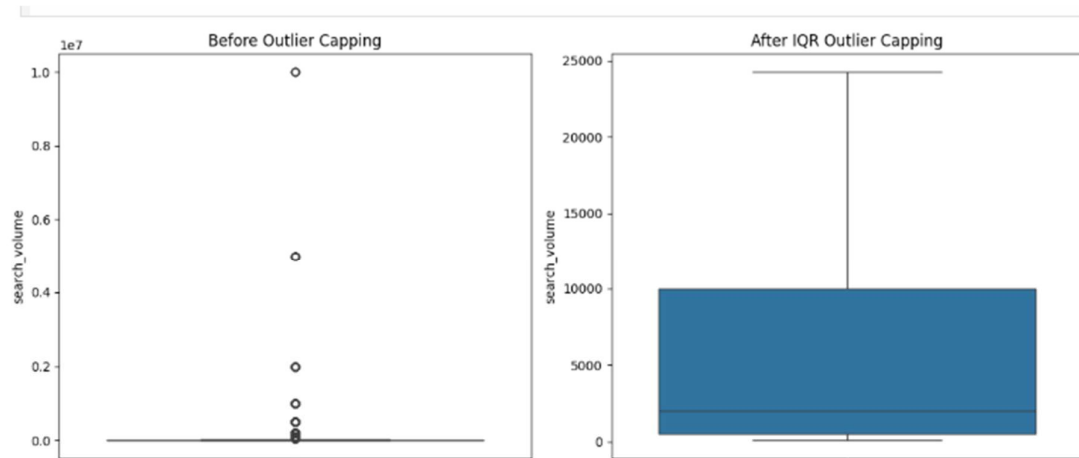
Outlier detection and IQR capping completed.

Initial Data Types:

```
query          object
start_date     object
end_date       object
active         bool
search_volume  float64
increase_percentage float64
categories     object
trend_breakdown object
dtype: object
```

Updated Data Types:

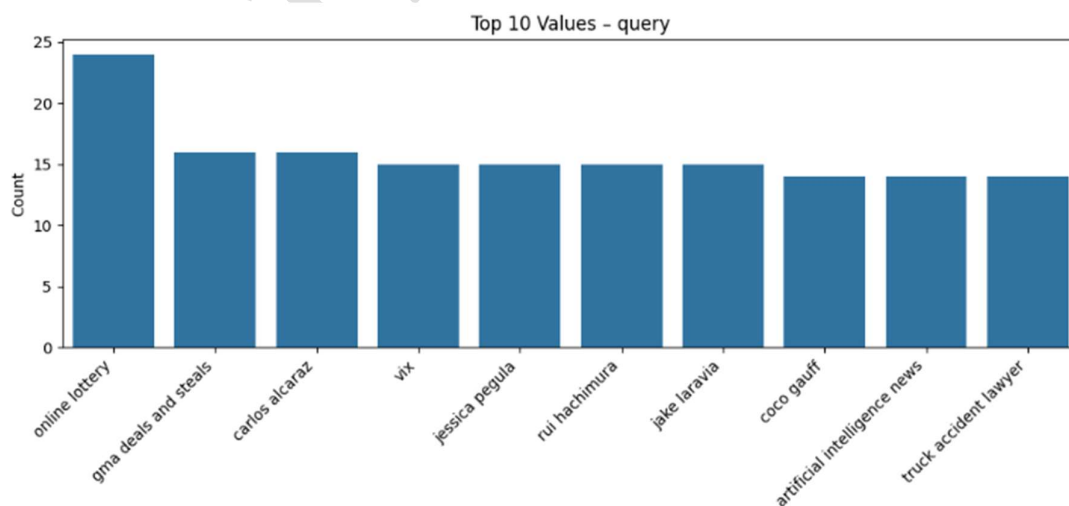
```
query          object
start_date     datetime64[ns, UTC]
end_date       datetime64[ns, UTC]
active         bool
search_volume  float64
increase_percentage float64
categories     object
trend_breakdown object
dtype: object
```



Text columns being normalized: ['query', 'categories', 'trend_breakdown']

Formatted text columns sample:

	query	categories	trend_breakdown
0	shedeur sanders	sports	dallas stars
1	whoopi goldberg	politics	america first award, mike johnson
2	mo williams	sports	mason williams basketball, mason williams
3	sloane stephens	sports, health	dallas stars
4	canceled tv shows 2026	entertainment	renewed and cancelled tv shows 2026



Final cleaned dataset shape: (31804, 8)

Remaining missing values:

```
query          0
start_date     0
end_date       0
active         0
search_volume  0
increase_percentage 0
categories     0
trend_breakdown 0
dtype: int64
```

Final data types:

```
query          object
start_date     datetime64[ns, UTC]
end_date       datetime64[ns, UTC]
active         bool
search_volume  float64
increase_percentage float64
categories     object
trend_breakdown object
dtype: object
```

First 5 rows of cleaned dataset:

	query	start_date	end_date	active	search_volume	increase_percentage	categories	trend_breakdown
0	shedeur sanders	2026-03-30 18:40:00+00:00	2026-03-19 10:00:00+00:00	True	200.0	50.0	sports	dallas stars
1	whoopi goldberg	2026-03-30 18:40:00+00:00	2026-03-19 10:00:00+00:00	True	5000.0	50.0	politics	america first award, mike johnson
2	mo williams	2026-03-30 18:00:00+00:00	2026-03-19 10:00:00+00:00	True	5000.0	300.0	sports	mason williams basketball, mason williams
3	sloane stephens	2026-03-30 17:50:00+00:00	2026-03-19 10:00:00+00:00	True	1000.0	200.0	sports, health	dallas stars
4	canceled tv shows 2026	2026-03-30 17:50:00+00:00	2026-03-19 10:00:00+00:00	True	5000.0	200.0	entertainment	renewed and cancelled tv shows 2026

Result: